

Predicting Asthma Exacerbations using Machine Learning

Abstract— Asthma exacerbations, characterized by sudden worsening of asthma symptoms, pose significant health risks and often require urgent medical intervention. Accurate prediction of these exacerbations can significantly enhance patient care and management. This study explores the application of machine learning, specifically the k-Nearest Neighbors (k-NN) algorithm, in predicting asthma exacerbations. Proposed method utilized a comprehensive dataset containing patient demographics, clinical history, environmental factors, and recent symptom data. The k-NN algorithm was trained and tested on this dataset to identify patterns and predict exacerbation events. The results demonstrate that k-NN, with appropriate parameter tuning and feature selection, achieves a promising prediction accuracy. This approach holds potential for integration into clinical decision support systems, offering a proactive tool for asthma management and reducing the burden of exacerbations on patients and healthcare systems. Future work will focus on enhancing model performance through the incorporation of real-time data and expanding the dataset to include diverse populations.

Keywords— Artificial Intelligence, Asthma exacerbation, k-NN, Machine Learning.

I. INTRODUCTION

Asthma is a chronic respiratory disease that affects millions of people worldwide. Symptoms of asthma are wheezing, coughing, and shortness of breath [1]. Asthma exacerbations can be very severe or life-threatening and requires immediate medical attention. Identifying patients at high risk of asthma exacerbations is very much important for effective disease management and prevention of hospitalization and severity. The dataset is taken from kaggle. It contains data of 300 patients with 9 features.

The aim of this proposed method is to predict the chances of Asthma in patients based on various factors such as age, gender, smoking status, medication, peak flow, and intensity of cough. This helps in early prediction and prevention of asthma. This also help doctors and medical professionals to make decisions and provide better care to their patients. The main aim or goal of the is to develop a k-NN model to predict the Asthma. The performance of the k-NN model will be compared with other machine learning models like decision tree to determine which is better approach for predicting asthma exacerbations in patients.

This paper has been organized as follows. Section II describes the Literature Survey undergone for the process. Section III explains the methodology used for the implementation of this proposed method. Section IV comprises of results and discussion, performance of different algorithms and comparisons between the algorithms used, it also includes various graphical representations for efficient analysis. Finally, Section V gives the conclusion of the proposed method.

II. LITERATURE SURVEY

[1] This paper explains the usage of different machine learning (ML) models used in predicting asthma exacerbations. In this research, 23 ML-based models were developed using various machine learning algorithms such as logistic regression, random forest, and boosting. Based on the performance metrics the best model is selected. The study also found that the performance is better with larger datasets. The results suggests that ML methods can be used to identify patients who are at risk and thus helping the medical professionals to take better decisions and provide better care to their patients.

[2] This study explains how mobile health technologies can be used in managing asthma. Machine learning can use mobile health data to provide personalized recommendations. In this, 22 different research papers which used mobile health technologies for asthma management was studied, categorizing them into three areas: developing new monitoring tools, predicting asthma attacks, and identifying subgroups of asthma patients. This study showed that machine learning can be used for asthma management. But most studies lacked real-world testing. The models should be tested in real world to analyze its correctness. This will help the doctors and patients managing asthma.

[3] This study presents asthma prediction among children using ML. In this study two prediction models were used: one for early life (CAPE) and one for preschool age (CAPP). The dataset was collected from the Isle of Wight Birth Cohort. They used seven different ML models for prediction. All these models were evaluated using performance metrics and it was found that Support Vector Machines (SVM) had highest accuracy. The CAPE model used data up to age 2 and the CAPP model used data up to age 4. CAPP model performed better compared to CAPE model with 82% accuracy. This study explains the usage of different ML models to predict asthma in children. This helps those children who are at risk.

[4] This work introduces a new method, called the Affinity Graph Enhanced Classifier (AGEC) which uses blood samples to predict asthma. The dataset consists information of 152 people with asthma. It was found that AGECEC was better at predicting asthma compared to other methods with an accuracy of 72.50%. AGECEC uses graphs make more accurate predictions by finding. It uses specific blood test results to make better predictions. This method helps the doctors to diagnose asthma more accurately and thus helps them to provide better health care to their patients.

[5] In this paper different machine learning models is applied to predict asthma development in children. The 2016 National Survey of Children's Health dataset is used here. The study shows the importance of features like sex, difficulty breathing, allergies, and medication in asthma prediction. It was found that among various classifiers random forest classifier was more effective in prediction which showed the accuracy of 90.9%.The paper tells that more work is needed to make accurate predictions which includes looking at more

datasets, verifying different machine learning algorithms and improving models for better prediction. It also tells the importance of early asthma detection in children. It aims to provide better care to children with asthma.

[6] The paper presents the usage of machine learning to predict childhood asthma. The dataset was collected from the National Survey of Children's Health. Different algorithms like linear regression, decision trees, random forests, k-NN and Naive Bayes was tested using this dataset. It was found that random forest model achieved the highest accuracy at 90.49%. The important factors used for predicting asthma were sex, difficulty breathing, allergies and medication use. Using this model the doctors can predict asthma in children and provide better care to children.

[7] This research paper tells about the work that is related to the prediction of asthma in a child. It predicts whether a child has asthma or not by using different machine learning algorithms. It uses algorithms such as support vector machines(SVM), decision trees and random forests. Based on the performance metric, obtained by using different algorithms the best model is selected. The study tells how imbalanced datasets can be handled. The predictions made are very much accurate. The different metrics used to evaluate the performance of these models are accuracy, sensitivity, and specificity. The paper highlights the importance of testing these models in real-world and ensures their effectiveness in predicting asthma in children.

[8] This Research paper tells how the researchers have developed a model to predict chronic respiratory diseases like asthma, COPD and lung cancer. This is done by studying tiny particles in the blood. The tiny particles are called microbial extracellular vesicles or in short they are also called as EVs. To develop this model they used machine learning algorithms. The model developed accurately predicts the chances or risk of being attacked by these diseases. The model has achieved the accuracy of 93% to 99%. This model has helped the doctors and medical professionals. Also it helps in early prediction and prevention of the disease. The researchers also tested their models and discovered the specific vitamins that could lower the risk of asthma and lung cancer. This study implies that analyzing EVs in the blood could be a effective way to diagnose and treat respiratory diseases.

[9] In this research paper we get to know how the researchers have used different machine learning models to diagnose asthma and COPD using a dataset. The dataset consists data of 132 patients. They tried with different machine learning algorithms like Random Forest, Naive Bayes, and Logistic Regression for developing this model. Among all the models, Random Forest has worked the best, with high precision rates of 80.3% for asthma and 97.7% for COPD. This study also identifies the important factors for the prediction such as smoking status or smoking history, lung function and age groups. This proposed method tells how machine learning can help in predicting the respiratory diseases especially asthma and COPD.

[10] This research paper tells how the researchers have used machine learning algorithm to predict asthma attacks. They have combined two algorithms, Support Vector Machine and Multilayer Perceptron to develop the model. The model analyses the patient's dataset to make predictions. Support Vector Machine has achieved the accuracy of 98.5% while Multilayer Perceptron has achieved the accuracy or

success rate of 90.5%. This model considers various factors like smoking history or smoking status, lung function, and age of the patients. These factors are used to find patterns in the data. This model tells how machine learning can be used in predicting or diagnosing the chronic respiratory diseases like asthma which can be life threatening if it becomes severe.

[11] This research is mainly about designing a smartwatch. The watch is developed such that it warns asthma patients about the air pollution. The researchers have used a computer program to analyze various features of the dataset. The features are like air, temperature, and humidity. This program predicts how these factors affect the asthma patients. The researchers want to design a smartwatch with a sensor that can detect these air pollutants and warn the asthma patients when their health might be at risk.

[12] In this paper a study was conducted using Random forest as a machine learning model in self-management of asthma. The authors validate their proposed techniques using the Asthma Mobile Health Study (AMHS) data, demonstrating that probabilistic and discriminant classifiers can provide early warning to asthma symptoms with high accuracy (AUC > 0.87). The research illustrates how machine learning can be used to personalize patient insights and care within the context of asthma management. Some additional directions that the authors recommend future work take are, using different types of data, i.e. weather, patient demographic information, parallel to the model that was done.

[13] This research paper examines how machine learning techniques can enhance the diagnosis and treatment of bronchial asthma. The authors used a deep learning method called convolutional neural networks (CNNs) to study a group of adult patients with bronchial asthma. According to the findings, the CNN model demonstrated superior performance compared to conventional approaches, accurately identifying asthma cases with high precision and recall rates. As a result, the authors propose that this technology has the potential to create tailored and efficient treatment strategies for individuals with asthma, ultimately resulting in better health results and decreased medical expenses.

Machine learning is used for the various recommending systems [14] and future work of the proposed system is to recommend the proper care of the health and excise.

III. METHODOLOGY

In this proposed method, preprocessing the data was a major step to ensure if it is suitable for machine learning analysis. The dataset was obtained from Kaggle [15] which is refined as per the standard instructions of doctors. Fig.1 shows the glimpse of the dataset which is used for predicting asthma exacerbations which includes various factors like gender, age, Smoking_Status, peakflow, medication, and intensity of cough.

The LabelEncoder library in Python to convert categorical variables into numerical values is used as it is necessary because many machine learning algorithms require numerical inputs to function effectively. The proposed system applies the k-NN algorithm to predict the chances of asthma in patients based on 9 various features. It is a machine learning algorithm which works by considering the nearest data points around it in the training dataset to make predictions. It classifies the new instances by considering the 'k' closest instances in the training dataset. The Euclidean distance method is used to find

similarity among the instances of the dataset. k-NN algorithm is implemented using scikit-learn library in python. The final model achieved an accuracy of 93% on the test dataset after continuously training the model using training dataset. By analyzing the performance metrics the k-NN is the best machine learning model for predicting asthma diagnoses compared to other algorithms.

k-NN has its own benefits over other algorithms. Firstly, it is straightforward to implement as it requires simple model tuning. This simplicity allows for quick deployment using standard libraries like scikit-learn. Secondly, k-NN is a non-parametric algorithm. This characteristic is particularly beneficial for datasets where relationships between features and the target variable are complex or non-linear. Thirdly, k-NN is relatively robust to noise and outliers, as it classifies based on multiple neighboring data points rather than relying on a single point. Additionally, k-NN's versatility allows it to be used for both classification and regression tasks, making it a flexible tool for variety of applications. k-NN is particularly well-suited for smaller datasets, as it does not require a large amount of data to train the model effectively. This attribute was especially useful for the proposed method, given the limited size of dataset. Finally, the simplicity of k-NN makes it easy to interpret and understand, which is crucial for explaining how the model arrives at its decisions.

Initially, the dataset was loaded into a pandas Data Frame. The preprocessing phase involved converting categorical variables into numerical values using the LabelEncoder, ensuring that the data was in a format suitable for machine learning algorithms. After preprocessing, the relevant features and the target variable is selected and stored in separate DataFrames. The data is split the data into training and testing sets using the train_test_split function from scikit-learn. This step was essential to assess the performance of the model on unseen data. The proposed system initialized the k-NN model with a specific number of neighbors, which was set to 3 based on our cross-validation and grid search results. The training phase involved fitting the k-NN model to the training data using the fit function. After training, the model was used to predict the target variable for the testing data using the predict function. To evaluate the performance of the k-NN model, The proposed system used several metrics, including accuracy, precision, recall, F1-score, sensitivity, specificity and confusion matrix. To assess the predictive capabilities of proposed model and optimize its performance, employed a range of metrics. Using these metrics, obtained the complete review of the predictive capabilities of the model and helped in fine-tuning the model. To implement this proposed method, used different tools and libraries that helped us to program and manipulate data using Python, perform data manipulation and analysis using pandas, implement the k-NN algorithm and other machine learning functionalities with scikit-learn, and convert categorical variables into numerical values using LabelEncoder.

Fig.2.shows the methodology which is used to implement our proposed method.

Patient_ID	Age	Gender	Smoking_Status	Medvalue	Medication	Peak_Flow	Intensity of cough	Asthma_Diagnos
1	26	Female	Non-Smoker	1	Inhaler	175	high cough	Yes
2	52	Female	Ex-Smoker	0	None	156	low cough	No
3	56	Female	Ex-Smoker	1	Inhaler	236	high cough	Yes
4	55	Male	Current Smoker	2	Controller Medication	378	medium cough	Yes
5	43	Female	Current Smoker	0	None	159	low cough	No
6	23	Male	Non-Smoker	0	None	376	low cough	No
7	33	Female	Current Smoker	0	None	296	low cough	No

Fig.1.Dataset used for Predicting Asthma Exacerbations

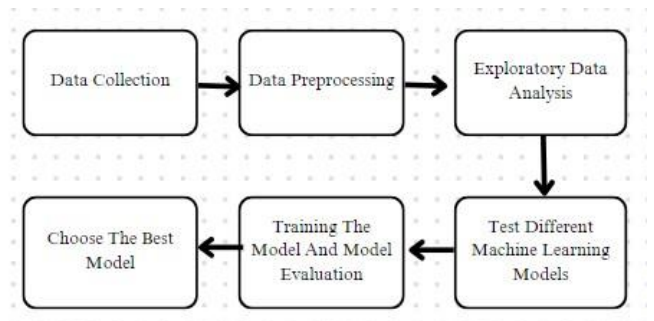


Fig.2.Methodology used in implementing the proposed method

IV. RESULT

Table 1: comparison of K-NN and decision tree based on the different metrics.

Algorithm	KNN	Decision tree
Accuracy	0.933333333333	1.0
Precision	0.9787234042553191	1.0
Recall	0.9019607843137255	1.0
F1-score	0.9387755102040817	1.0
Sensitivity	0.9019607843137255	1.0
Specificity	0.9743589743589743	1.0
Mean Absolute Error(MAE)	0.06666666666666667	0.0
Mean Squared Error(MSE)	0.06666666666666667	0.0
R-Squared Error	0.7285067873303168	1.0

The obtained results when the dataset is tested using k-NN and decision tree model is shown in table 1. The decision tree model is overfitting as it has 100% accuracy, which means that it is performing well on the training data but when it comes to test data the model does not perform well or it shows poor results. This maybe because the decision tree is too complex and it memorizes the training data rather than learning patterns in the dataset. So tried using the k-NN. This model does not result in overfitting and performs well on small dataset. This is because k-NN model is a simpler model and it has less parameters to adjust. Here are some reasons why the k-NN model is a better choice in this case:

- **Less Overfitting:** The k-NN model is less prone to overfitting because it does not have as many parameters to adjust. This makes it more robust to changes in the data and less likely to memorize the training data.
- **More Generalizable:** The k-NN model is more generalizable because it is based on the idea of finding the most similar samples in the training

data. This makes it more likely to perform well on new, unseen data.

- **Easier to Interpret:** The k-NN model is easier to interpret because it is based on the idea of finding the most similar samples in the training data. This makes it easier to understand why the model is making certain predictions.
- **Less Sensitive to Hyperparameters:** The k-NN model is less sensitive to hyperparameters because it does not have as many parameters to adjust. This makes it easier to tune the model and less likely to overfit.
- **Better Handling of Noise:** The k-NN model is better at handling noisy data because it is based on the idea of finding the most similar samples in the training data. This makes it more robust to outliers and noisy data.

k-NN and Decision Tree performance measures is shown in fig.3. The information includes mean absolute error (MAE), mean squared error (MSE), recall, accuracy, precision, sensitivity, specificity, F1-score, mean squared error, and R-squared error. Each metric assesses several parts of the model's performance according to its ability to manage both positive and negative aspects, produce accurate forecasts, and quantify prediction mistakes.

In this instance, Decision Tree is showcasing its accuracy in classifying and predicting errors without any mistakes. k-NN algorithm, on the other hand, has somewhat lower values in these metrics. While Decision Tree performs exceptionally well across all assessed measures, k-NN offers good overall performance, making Decision Tree a more effective algorithm for the given dataset.

Fig.3. is a bar chart representing the distribution of 'medvalue' across different 'Age' groups. But, by looking at the graph conclude that there is no potential relationship between the two variables.

Fig.4. is a bar chart representing the 'medvalue' plotted against 'intensity of cough'. It shows the distribution of 'medvalue' across different levels of intensity of cough. This graph visualizes the relationship between cough intensity and 'medvalue'. But more work needs to be done to analyze the strength and direction of this relationship.

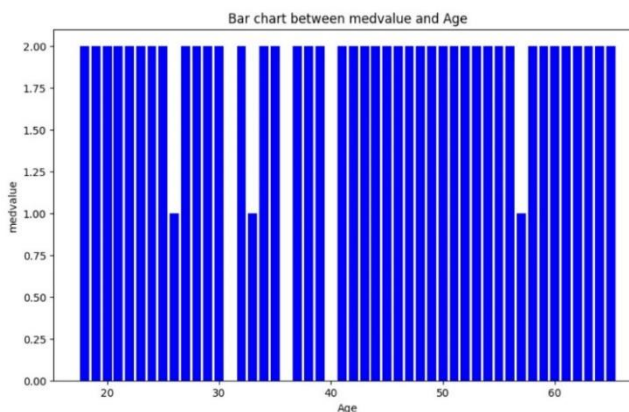


Fig.3.Barchart of medvalue vs Age

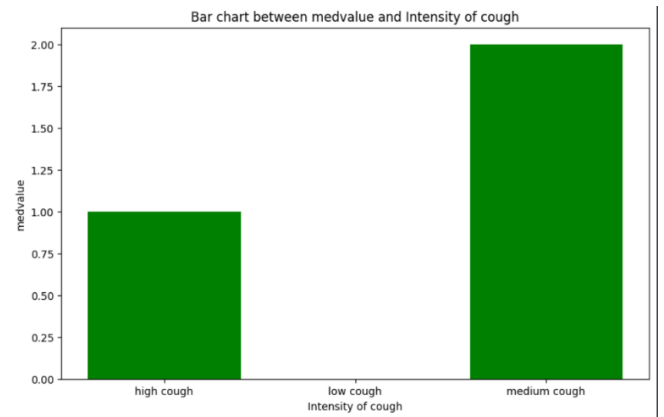


Fig.4..Bar chart of medvalue vs intensity of cough

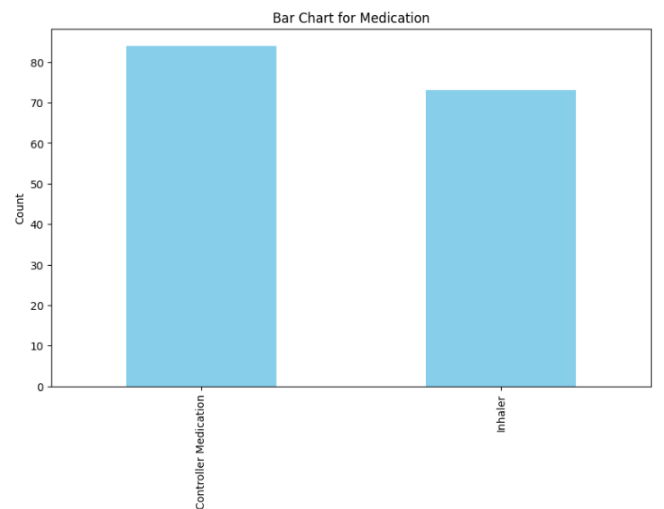


Fig.5.Bar chart for frequency of type of medication

Fig.5.shows the bar chart that depicts the frequency of each type of medication present in the dataset, providing insight into the distribution of medication usage among the observations. It indicates that certain medications are more prevalent than others within the dataset.

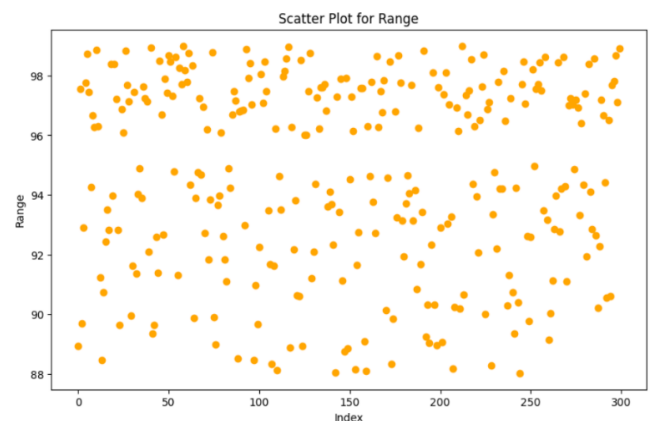


Fig.6.Scatter Plot which shows the distribution of Range vs Index

Fig.6 shows the scatter plot that visualizes the distribution of the 'Range' variable across the dataset's index, indicating the variability of 'Range' values throughout the dataset. It

suggests potential patterns or outliers in the 'Range' variable's distribution.

Fig. 7 shows the graphical representation of the final result on comparing k-NN and decision tree.

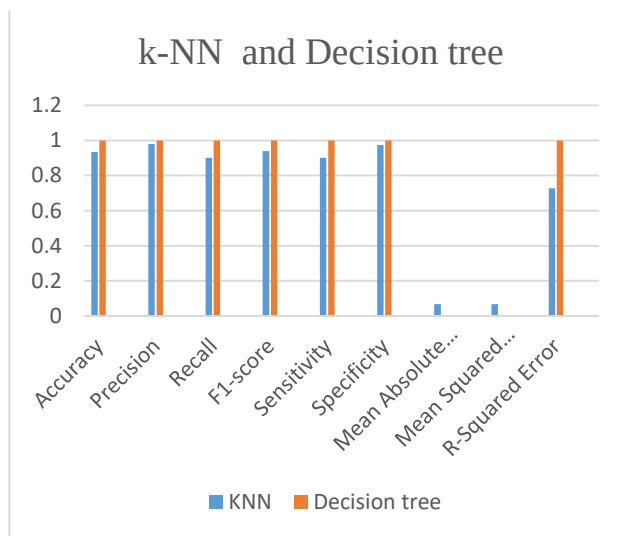


Fig.7.k-NN and Decision Tree

V. CONCLUSION

The proposed method is dived into machine learning to predict asthma exacerbations and the dataset we used is from Kaggle. One of the most promising tools found was the k-NN model using which it is possible to strike an accuracy of 88% with the data the model had never seen before; this suggests an opportunity to identify patients at risk of subsequent flares earlier, for intervening and improved health outcomes. The optimal decision tree model outcome seems nearly perfect, but this is an example of overfitting. Too much of a fit to training data makes it harder for a model to generalize and predict accurately for new patients. The importance of training and testing data splits: A model can do really well if it memorizes the specific examples, but it is only useful when it generalizes, i.e., learns the overall patterns and trends of the data. Altogether, this work provides an important groundwork for future investigation in this area. With a model built to include more salient characteristics of asthma exacerbations, paired with different algorithms not as biased toward overfitting, and seeing a similar validation with a bigger pool of data might be one step closer to developing a model that can be used in practice to predict asthma exacerbations early on stop. In the long run, this can lead to a great improvement in patient care and quality of life as well. The future progress of developing a practical tool to predict your next asthma exacerbation may be just around the corner but may involve some combination of incorporating more clinically diagnostic characteristics of asthma exacerbations using less-overfitting dependent algorithms, via validation in a larger dataset. These advances could provide tangible benefits in terms of patient care and quality of life.

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